

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

Chemistry — SSC-I (9th Grade)

Syllabus: Chapter 12 — Environmental Chemistry: Water**Total Marks: 65****Time Allowed: 2h 30m****Version: 1****General Instructions:**

1. Write your name, roll number and section clearly on the answer sheet.
2. Section A is compulsory. Encircle the correct option on the question paper itself.
3. In Section B, attempt **all 11** questions. For each question, attempt either the main question **OR** its alternative.
4. In Section C, attempt **all 4** questions. For each question, attempt either the main question **OR** its alternative.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Question	A	B	C	D	Answer
1	Anhydrous copper(II) sulphate is used to detect water because it turns:	yellow	blue	green	red	A B C D
2	The melting point of pure water is:	0°C	4°C	100°C	37°C	A B C D
3	The pH of distilled water is:	Less than 7	Greater than 7	7	None of these	A B C D
4	Which substance is added during the coagulation step of water treatment?	Chlorine	Slaked lime and alum	Sand and gravel	Activated carbon	A B C D
5	Chlorination of water produces hypochlorous acid, which kills:	Heavy metals	Bacteria	Phosphates	Nitrates	A B C D
6	Which waterborne disease is caused by the parasite <i>entamoeba</i> ?	Typhoid	Hepatitis	Dysentery	Jaundice	A B C D
7	Which of the following is NOT a waterborne disease?	Hepatitis	Typhoid	Dysentery	Anaemia	A B C D
8	High concentrations of phosphates in water bodies cause:	Typhoid	Eutrophication	Chlorination	Sedimentation	A B C D
9	NPK fertilizers provide which three elements to plants?	Na, P, K	N, P, K	N, Pb, K	N, P, Ca	A B C D
10	Which activity of humans results in contamination of water bodies?	Livestock waste	Pesticides	Septic tanks	All of these	A B C D
11	Distilled water is preferred in laboratories to prepare chemical reagents because:	it contains minerals	it is free of contaminants	it has high conductivity	its pH is greater than 7	A B C D
12	Water scarcity in Pakistan is mainly caused by:	Over-filtration	Population growth, climate change, and inadequate supply	Excess dissolved oxygen	Too many water treatment plants	A B C D

SECTION B — Short Questions (11 × 3 = 33 Marks)

Attempt all 11 questions. For each pair, attempt either the main question OR its alternative.

Part	Question	Marks	OR	OR Question	Marks
2(i)	Describe Activity 12.1. What colour change confirms the presence of water when anhydrous copper(II) sulphate is used? Write the equation for the reaction.	3	OR	How is the purity of water tested using its melting point? What does a melting point below 0°C indicate about a water sample?	3
2(ii)	State three physical properties of pure water. At what temperature does pure water boil and freeze at sea level?	3	OR	Explain why natural water such as rainwater and groundwater is not pure. Name four substances that water dissolves as it falls through the atmosphere.	3
2(iii)	Name any three substances found in natural water that are harmful. For each, state one specific harm it causes.	3	OR	Name two substances found in natural water that are beneficial, and explain why each is necessary for life.	3
2(iv)	List the four steps of raw water treatment in order. State what is removed or achieved at each step.	3	OR	Write the chemical equation for the coagulation step in water treatment. Name the gelatinous product formed and explain its role.	3
2(v)	Write the equation for chlorination of water. Name the acid produced and explain how it makes water safe to drink.	3	OR	Explain why chlorination is not effective against viruses such as those that cause hepatitis. What method is needed to treat	3

				viral contamination?	
2(vi)	Compare distilled water and tap water under: (a) source (b) purity (c) one application each in the chemistry laboratory.	3	OR	Why is long-term drinking of distilled water not advisable? What does the "Do You Know?" box in the textbook say about this?	3
2(vii)	Define water pollution. List any three human activities that cause water pollution.	3	OR	Explain how household waste contaminates water bodies. Name three infectious diseases that can spread through contaminated household water.	3
2(viii)	What is dysentery? Name the causative organism. How does it spread and what are its main symptoms?	3	OR	What is typhoid? Name the bacteria responsible. What is the characteristic fever range in typhoid, as stated in the textbook?	3
2(ix)	State three ways to deal with the negative effects of water pollution as given in Section 12.5 of the textbook.	3	OR	Why are heavy metals such as arsenic (As), chromium (Cr), and lead (Pb) dangerous water pollutants? Name two health problems they can cause.	3
2(x)	List three factors that cause water scarcity in Pakistan, as described in Section 12.6.	3	OR	Give any three measures recommended in the textbook to solve water scarcity in Pakistan.	3
2(xi)	Define fertilizers. State the three main nutrients provided by NPK fertilizers, giving one example of a fertilizer that supplies each nutrient.	3	OR	What is nutrient management? Explain how over-use of fertilizers causes environmental problems, specifically in water bodies.	3

SECTION C — Long Questions (4 × 5 = 20 Marks) — Attempt all 4

Part	Question	Marks	OR	OR Question	Marks
3(i)	(a) Describe how anhydrous copper(II) sulphate detects water. State its colour before and after the test. Write the equation for the reaction. [3] (b) Explain how the boiling point test is used to check the purity of water. What does a boiling point above 100°C indicate and why? [2]	5	OR	(a) Compare distilled water and tap water in detail: how each is obtained, three differences in properties, and two differences in their laboratory uses. [3] (b) Explain why natural spring water is better for drinking than distilled water. What essential substances does natural water provide that distilled water lacks? [2]	5
3(ii)	(a) Describe all four steps of the raw water treatment process in a municipal water purification plant. For each step, name the materials used and what is removed. [4] (b) Draw a flowchart showing the order of steps: Raw Water to Pure Water. [1]	5	OR	(a) Write the chemical equations for: (i) coagulation (using slaked lime and alum) and (ii) chlorination of water. Name the product in each that acts against bacteria or dirt. [3] (b) Explain why water treatment is essential for public health. State any three diseases prevented by proper water treatment, as mentioned in the textbook. [2]	5
3(iii)	(a) Describe four waterborne diseases covered in Section 12.4: for each, state the causative organism (or type), how it spreads, and one symptom. [4] (b) State one measure to prevent each: dysentery and typhoid. [1]	5	OR	(a) Identify three sources of water pollution described in Section 12.3 and explain the specific harm each causes to human health or aquatic life. [3] (b) Describe three ways to deal with the negative effects of water pollution as given in Section 12.5. [2]	5
3(iv)	(a) Explain water scarcity as a major problem in Pakistan. Describe any three causes and three solutions recommended in Section 12.6 of the textbook. [4] (b) How does climate change specifically worsen water scarcity in Pakistan? [1]	5	OR	(a) Define fertilizers. Explain what NPK fertilizers are and name two examples of fertilizers that supply nitrogen and two that supply phosphorus. [3] (b) Explain how excessive use of fertilizers causes eutrophication in water bodies. What is the final impact on aquatic life? [2]	5

Invigilator's Signature: _____

Candidate's Signature: _____

Chapters Covered: Chapter 12 — Environmental Chemistry: Water (pp. 163-173)

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